

Tutorial 7: Sept 10, 2025

(1) Thomas Finney.

problem 41 p 951

problem 5 p 950.

problems 17-22 p 967 (select any 2).

problem 8 on p 987.

(2) Notes: problem on p110. (if time permits).

(3) Notes: p 94 (if not done in tut6).

(4) Pictures: $x^2 + y^2 + z^2 = 1$ (Sphere).

$$x^2 + y^2 - z^2 = \alpha^2 \quad (\alpha \neq 0)$$

$$x^2 - y^2 - z^2 = \alpha^2 \quad \alpha \in \mathbb{R}.$$

$$x^2 + y^2 - z^2 = 0 \quad (\text{Cone})$$

$$z = y^2 + x^2 \quad (\text{paraboloid})$$

$$z = x^2 - y^2. \quad (\text{Hyperbolic paraboloid}).$$